What we compare against what, and why the vertex cap moves out of the RD figure

All numbers are measured runs. 13 Nov 2026.

1 Two questions, answered separately

- **Is the codec good?** Take the same meshes and change only the codec. Our files are 2.4 to 2.5 times smaller than TFAN's. This is the only comparison where nothing but the codec changes.
- **Is the whole system good?** In the RD figure, TFAN needs 4.1 to 8.2 times more bytes than we do for the same PSNR. This number is larger than the codec gain because it also includes the gain from training the mesh at the target size. It is a system comparison, not a codec comparison, and the paper will say so.
- **Where does the vertex cap go?** Out of the RD figure and into the ablation table. At the same file size we are 0.33 dB above the cap (better in 34 of 36 comparisons). A gap of 0.33 dB cannot be seen in a plot that spans tens of MB, which is why the orange curve did not look higher than the blue one. The cap row uses our codec and is trained at the target size, so it is our own system with one part swapped. It is an ablation, not an outside baseline.

2 The vertex cap does not decide the file size

Each line in Figure 1 is one scene. Every scene in a panel is plotted on the same cap axis.

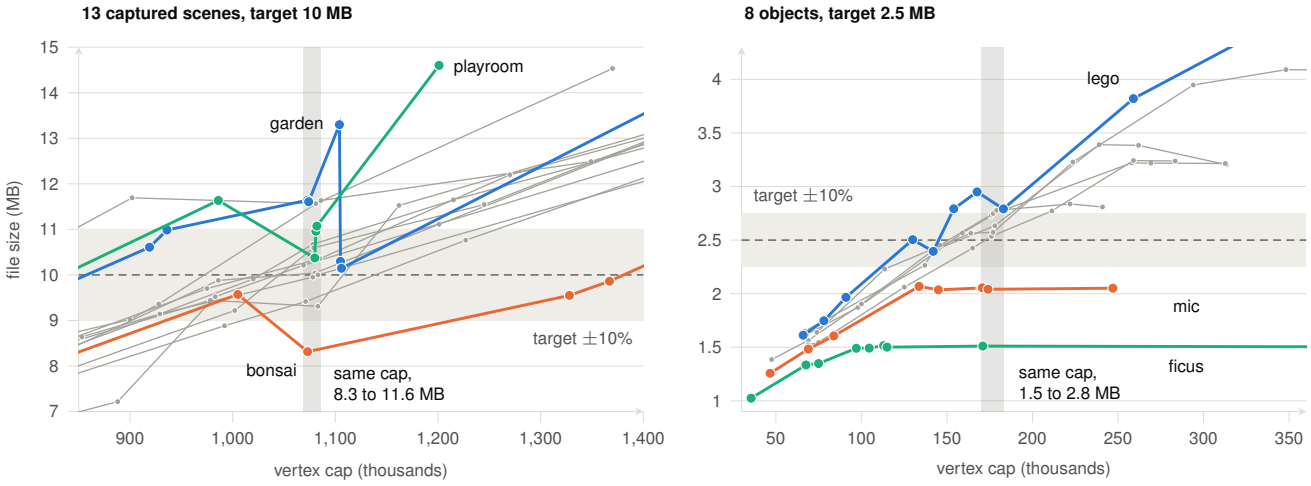


Figure 1: File size against vertex cap, one line per scene, every cap run in the plotted range. Three scenes per panel are coloured, the others are grey. Dashed line and light band: target and ±10%. Dark vertical strip: the cap at which every scene of the panel was trained.

What the figure shows:

- **Same cap, different scenes, different sizes.** With a cap of about 1.08 million vertices the 13 captured scenes give files from 8.3 to 11.6 MB. With a cap of about 176 thousand the 8 objects give files from 1.5 to 2.8 MB.
- **Same scene, almost the same cap, different sizes.** Garden: a cap of 1,104,000 gives 13.3 MB and a cap of 1,105,000 gives 10.3 MB.
- **A larger cap can give a smaller file.** Bonsai: 1,005,000 gives 9.6 MB, 1,073,000 gives 8.3 MB.
- **Raising the cap can do nothing.** Mic stays at about 2.05 MB from a cap of 134 thousand up to 247 thousand. Ficus stays at 1.5 MB from 97 thousand up to 515 thousand.

scene	cap	file (MB)	what happened
garden	1,104,000	13.30	
garden	1,105,000	10.30	cap +0.1%, file −23%
garden	1,106,000	10.15	
playroom	1,080,000	10.37	
playroom	1,081,000	10.96	cap +0.1%, file +6%
bonsai	1,005,000	9.57	
bonsai	1,073,000	8.32	cap +7%, file −13%
drjohnson	902,000	11.69	
drjohnson	1,081,000	11.57	cap +20%, file −1%
lego	130,000	2.50	
lego	142,000	2.40	cap +9%, file −4%
lego	154,000	2.79	cap +8%, file +17%
lego	183,000	2.79	cap +19%, file unchanged
mic	133,718	2.07	
mic	247,000	2.05	cap +85%, file −1%

Table 1: Examples from the figure, each row compared with the row above it.

Over all cap runs, counting pairs of runs on the same scene:

- caps less than 1% apart: 11 pairs. In 4 of them the larger cap gave the smaller file. In 2 the file changed by more than 10%.
- caps less than 5% apart: 29 pairs. In 12 the larger cap gave the smaller file. In 8 the file changed by more than 10%.

3 What this means for hitting a size

21 scenes, one target each: 10 MB for the 13 captured scenes, 2.5 MB for the 8 objects. “Ours” measures the real file every 500 training steps and removes whatever is above the target; one training per scene.

	Vertex cap	Ours
Cap chosen from	other scenes, same dataset	not needed
Within ±10% after 1 training	15 / 21	20 / 21
Within ±10% after 3 trainings	18 / 21	–
Within ±2% after 1 training	3 / 21	16 / 21
Never within ±10%	drjohnson, mic, ficus	ficus
PSNR at the same file size	reference	+0.33 dB (34 of 36)

The cap baseline was given its best case: a cap taken from scenes of the same dataset and up to three trainings. We propose to stop here. Any further variant of the cap baseline would be a new method that we design ourselves, so its result could always be questioned.